

2.11 Duct Insulation

- a) All insulation materials shall be Class – O as per BS 476.
- b) All air conditioning ductwork shall be insulated with closed cell elastomeric insulation.
- c) All ductwork exposed to weather shall be insulated with two layer of 7mm thick fiber glass cloth plus two layers of shield coating.
- d) The insulating material shall have a K factor of not more than 0.0332 W/degk at 24°C mean temperature. The insulation thickness shall be 15 mm for supply and 13 mm for return ducts and shall be of 40 - 60 kg/m³ nominal density. For return duct located below the roof and pass through non-air conditioned areas, minimum insulation thickness shall be provided.
- e) All duct surfaces shall be cleaned and given one coat of flinkote before the application of the insulation material. A further layer of flinkote shall be applied on the insulation before they are wrapped with the vapour barrier which shall have a minimum of 75 mm overlap at all joints and securely sealed with cover strips and adhesive duct tape to form a perfect vapour barrier. Anti-corrosion treated steel tape shall be strapped around the insulated ductwork at 1 m centres to prevent the insulation from sagging.
- f) Where indicated in the Tender Drawings, the materials used for internal insulation(acoustic) shall be open cell elastomeric type. Class '1' non-flammable acoustic insulation of 15 mm (minimum) thick and 140-180 kg/cum density. All the internal insulation material shall be securely fastened to the ductwork using perforated aluminium sheet if necessary. The material shall be self-extinguishing to ASTM635 and has a thermal conductivity of not more than 0.0465 w/mK at 25°C mean temperature. In addition, a flame retardant glass cloth layer shall be provided for all internal insulation used.
- g) The internal insulation shall be supplied around the inside of the duct such that the insulation in continuous. The insulation and the duct corners shall be either butt joint or mitred and the aluminium lamination shall overlap but not less than 10 mm and glued securely by an approved adhesive. The studs shall be brazed to the ducts at spacing specified hereinafter, and shall have a length of not more than 1½ times the thickness of the internal insulation which it holds.
- h) The clips shall be galvanised and of the self-locking type of not more than 25 x 25 mm in size. The centre to centre spacing of studs shall be as follows:

<u>Duct Wall</u>	<u>Centre Spacing (mm)</u>
Top Side	150 mm
Vertical Side	200 mm
Bottom Side	300 mm

- i) Internally line ducts shall be inspected before installation. Duct walls shall not be punctured when securing the internal insulation.
- j) Where the internal insulation terminates, the insulation shall be tapered in ratio of 1 in 3. The aluminium lamination shall extend, over by 100 mm and taped securely to the duct. An overlap of 300 mm (12") shall be provided where the insulation changes from internal to external. All joints of the insulation material should be further covered with 50 to 100 mm.
- k) The supply ducts leaving air handling units and return duct to the AHU shall be lined internally for a minimum distance of 5 m.

Comply with deviation
k value 0.032 w/mk at 20 C

Comply with deviation K
value 0.047 w/mk at 20 C

Noted



ACOUSTIC INSULATION OF SHEET METAL DUCTING

- a) This section details the supply and application of Acoustic insulation to the sheet metal ducting, in accordance with these specifications and general arrangement shown on the drawings.
- b) The scope of the contractor with supply & installation of material shall include but not limited to the following items.
 1. Insulation materials of all types as specified and required.
 2. Insulation adhesives vapour barriers and finishing materials of all types as specified and required.
 3. Auxiliary materials such as binding and lacing wires, wire netting, bands, screws, pop rivets etc., as specified and required.
 4. Angles, clamps, lugs etc., for supporting insulation.
 5. RP Tissue, Aluminium perforated sheet.
 6. Any other material as may be required for making the insulation complete.
- c) All materials shall be new and fresh, incombustible, fire retardant, rot-proof, non-hygroscopic, venin proof, fungus proof, non-injurious to health, chemically inert, non-corrosive to steel and aluminium (even if soaked in water for extended periods), environment friendly and shall be guaranteed to withstand continuously and without deterioration the minimum and maximum temperatures to which these shall be subjected to under the specified applications. The insulation materials and any component of the finished insulation job shall not react chemically, singly or in combination, with water or moisture to form substances that are more actively corrosive to applied surface than water or moisture alone.
- d) The insulation material shall be kept dry at all times during transport, storage and installation. Decking and covering tarpaulins alone are not adequate for any length of time and shall not be allowed except in extreme emergencies and only for short period. Stacking of insulation material directly on ground shall not be done. No wet insulation shall be installed. If wet insulation is present, it shall be removed and replace with new dry insulation. The insulation may be protected with plastic film but shall be vented to prevent sweating.
- e) In order to protect the workers from the hazards of insulation materials, suitable protective gadgets shall be provided. Required safety precautions shall be taken during handling and application of insulation.
- f) **MATERIAL SPECIFICATION**

Duct liner (Insulation) of specified thickness should have the following properties and sound absorption co-efficient as per ASTM E 84 test & which should also meet NFPA 90A supplementary materials for air distribution system.

- Max Nominal Thermal conductivity at 20°C mean temperature shall be 0.037W/m.°C., as per ASTM C518, ASTM C177.
- Working Temperature Limitation shall be up to 121°C (as per ASTM C411) Alkalinity shall be PH 9.
- Should not cause or accelerate Corrosion & Odour Emissions., as per ASTM C665.
- Should not allow any Microbial growth, as per ASTM D 2020, ASTM G 21-Fungus, ASTM G 22-Bacterial.
- Moisture absorption should be less than 0.2% by volume in condition of 95% relative humidity for 96hours as per ASTM C 553.
- Flame spread index should be 0-25 and the smoke spread index should be 0-50., as per ASTM E84, UL 723.

Noted

Comply



Sound Absorption Coefficients								
Fibreglass density in kg/m ³	Fibreglass thickness in mm.	125 Hz	250 Hz	500 Hz	1 kHz	2 kHz	4 kHz	NRC
32	25	0.25	0.6	0.87	0.88	0.88	0.93	0.8
32	40	0.24	0.66	1.04	1	1.03	1	0.95
32	50	0.29	0.72	1.13	1.06	1	1.02	1
48	25	0.34	0.52	0.73	0.96	1.1	1.09	0.85
48	40	0.25	0.72	1.05	1.04	1.02	1.08	0.95
48	50	0.31	0.81	1.16	1.09	1.06	1.13	1.05

Noted

g) ACOUSTIC INSULATION OF THE MECHANICAL ROOM

- The walls & ceiling of the room to be cleaned.
- The frame work of 1mm. thick GI fabricated channels of 25x50mm size screwed back to back at 600mm centres shall be provided vertically and horizontally so that 600x600mm squares are formed.
- Wall Acoustic Insulation of Mechanical rooms / AHU rooms shall be done with GI frame work using 1mm thick formed channel and with 25 mm thick open cell insulation of density not less than 140kg/m³. 25 mm wide GI frame work of grid size 1200 mm x 900 mm shall be considered to hold the insulation material firmly to AHU wall.
- The same shall be finished with 22G aluminium perforated sheet of 30 to 40% perforation, fixed with sheet metal screws over the GI frame work. Screws shall be provided uniformly at regular intervals not more than 100mm. c/c. Overlapping of aluminium sheet shall be covered with aluminium beading.

h) ACOUSTIC INSULATION:

- Material shall be Engineered Nitrile Rubber open cell foam.
- The Random Incidence Sound Absorption Coefficients (RISACs) across the octave band frequencies; tested as per ISO 354, and Noise Reduction Coefficient (NRC) for the Acoustic Insulation thicknesses of 30mm and 50mm should be minimum as per the below chart:

Freq (Hz)	125	250	500	1000	2000	4000	NRC
30 mm	0.07	0.32	0.99	1.16	0.93	1.08	0.85
50 mm	0.23	0.73	1.29	0.99	1.09	1.11	1.05

Noted

- The material should be fibre free.
- The density of the acoustic insulation should be minimum 140 Kg/m³.
- All four walls and ceiling of air conditioning plant room and air handling unit rooms should be provided with acoustic lining.
- The insulation should have Built-in Anti-Microbial Product Protection, and should pass Fungi Resistance as per ASTM G 21 and Bacterial Resistance as per ASTM E 2180.
- The material should have a thermal conductivity not exceeding 0.047 W/m.K @ 20 Deg. C.
- The material should withstand maximum surface temperature of +850C and minimum surface temperature of -200C.



- The material should confirm to Class 1 rating for surface spread of Flame in accordance to BS 476 Part 7 & UL 94 (HBF, HF 1 & HF 2) in accordance to UL 94, 1996.
- The acoustic insulation should be tested and approved by Sound Research Laboratories Ltd., U.K.
- The insulation should be installed as per manufacturer's recommendation.

- Exhaust Systems Extracting Cold Air

- l) Where the Tender Drawings indicate that an exhaust system is extracting a portion or full amount of cold air (27°C and below) from an air-conditioning space, that part of the exhaust system, i.e. from the point(s) of extraction to the exit of the exhaust fan shall be externally insulated with closed cell elastomeric insulation
- m) The thickness of insulation shall be 15 mm thick and 40 - 60 kg/cu m density. Insulation details shall be similar to air conditioning duct.
- n) Where the Tender Drawings indicate a ceiling plenum return of the cold air, the fresh air supply system shall be insulated from the fan entry to the supply grille. The thickness of insulation and details shall be as above.

2.12 Cleaning of Ductwork

- a) During installation, properly fitted sheet metal covers shall be provided at all times to prevent ingress of rubbish. Rags, newspaper and like materials will not be accepted as a suitable covering.
- b) Should existence of rubbish be found in air ducts, the ductwork shall be dismantled as necessary to ensure complete removal of rubbish.
- c) All ducts shall be cleaned of dust before starting of fans.
- d) Duct Test Openings
- e) Openings shall be provided in ducts for testing purposes with a velometer or Pitot Tube on the supply and discharge side of air handling units major ducts and other necessary positions to be indicated to the Sub-Contractor in accordance with B.S. 848.
- f) The test openings are to be 150 mm centres across one side of the duct. Each opening shall consist of a 25 mm trap screw brazed into a 40 mm x 5 mm bar. The bar shall be rivetted to the duct across its width and each trap screw is to be fitted with a plug.
- g) Where test openings are fitted to internally insulated ducts, the duct insulation is to be protected with perforated metal internally for a distance of 600mm ahead of and 150mm past the test opening position.

2.13 Flexible Connections

- a) Flexible connection shall be provided at all ductwork connections to the suction and discharge openings of fans, FCUs, AHUs and where shown on the Drawings. These shall be made of approved flame retarded fabric or shall consist of sleeve joints with packing of non-combustible material of not less than 100 mm long.
- b) Flame retardant fabric connections shall be made of fibre glass cloth impregnated with a lead loaded vinyl. The surface density shall not be less than 5 kg/m². They shall be to the approval of the local fire authorities and the PMC.
- c) Flexible connections shall be so constructed to ensure that there is no air leakage. Connections shall have 50 mm to 100 mm length free of stress so as to permit renewal of

Noted



fabric without disturbing ductwork or other part of equipment.

- d) Flexible connection shall be measured by their cross section area perpendicular to the direction of air flow.

2.14 Access Doors

- a) The Sub-Contractor shall provide access doors on all ductwork connections, either upstream or downstream of coils, upstream of fans, fire dampers, filters etc. for inspection, cleaning and maintenance purposes in accordance with SMACNA recommendations.
- b) Access doors shall be of double skin construction with insulation of the same general character as adjacent sections of ductwork. The door panel shall be of not less than 1.0 mm sheet metal and shall have not less than 12 mm wide neoprene rubber gasket around their entire perimeter. It shall have the same fire ratings as the adjacent construction.
- c) Access doors shall be hung on heavy brass plated flat hinges and shall be secured in the closed position by means of wedge type catches. All hinges shall be so placed to enable the doors to remain open without additional support. All doors will be hinged with two hinges and two catches will be required on doors over 40 mm x 400 mm. Where it is impractical to use hinged doors, the access doors may be fixed in position with wedge type catches on two opposite sides, with a minimum of 4 catches per door.

3. MEASUREMENT OF INSULATION

Unless otherwise specified measurement for insulation for the project shall be on the basis of centerline measurements described herewith

- a. Pipe Insulation shall be measured in units of length along the centerline of the installed pipe, strictly on the same basis as the piping measurements described earlier. The linear measurements shall be taken before the application of the insulation. It may be noted that for piping measurement, all valves, and orifice plates and strainers are separately measurable by their number and size. It is to be clearly understood that for the insulation measurements, all these accessories including cladding, valves, orifice plates and strainers shall be considered strictly by linear measurements along the center line of pipes and no special rate shall be applicable for insulation of any accessories, fixtures or fittings whatsoever.
- b. Duct Insulation and Acoustic Lining shall be measured on the basis of surface area along the center line of insulation thickness. Thus the surface area of externally thermally insulated or acoustically lined be based on the perimeter comprising center line (of thickness of insulation) width and depth of the cross section of insulated or lined duct, multiplied by the center-line length including tapered pieces, bends, tees, branches, etc. as measured for bare ducting.

DATA TO BE FURNISHED BY THE CONTRACTOR AFTER THE AWARD OF CONTRACT

- a) List of drawings and documents to be submitted for review, approval and information with scheduled submission dates.
- b) Quality Assurance Plan (QAP)
- c) List of insulation materials with specifications and properties like density, thermal conductivity etc.,
- d) Values of thermal conductivity at various mean temperatures for all insulation materials.
- e) List of insulation adhesives, vapor barriers and finishing materials with specifications.
- f) List of auxiliary materials with specifications.
- g) Test certificates for insulation and other materials
- h) Detailed insulation application and finishing procedure.

Noted

